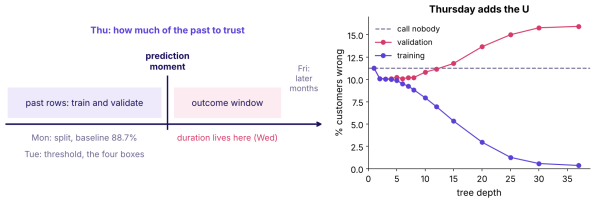
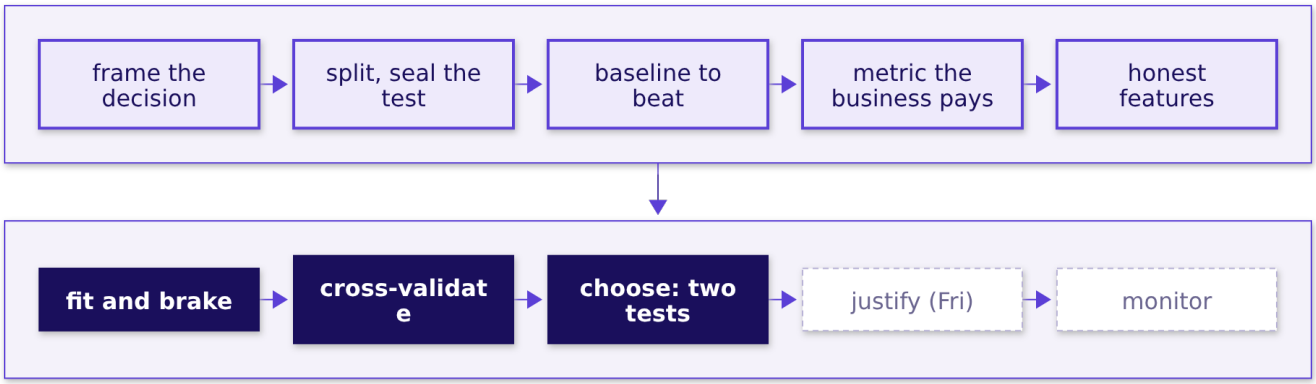
Cheat sheet: generalise or memorise

Week 5, Thursday. Redraw the picture from memory before you read the rest.



The ML process, today lit



Today's hard terms

Term	Plain meaning	Rule or formula	Trap
Overfitting (variance)	learned the training rows, surprised by new ones	gap = validation error minus training error	reading 99.6 on training as ready
Underfitting (bias)	too simple to be right anywhere	both errors high and close, near the baseline	blaming the data and adding rows
Learning curve	training and held-out error as rows grow	wide gap: brake; close and high: capacity	more rows do not close a gap made of noise

Term	Plain meaning	Rule or formula	Trap
Regularisation	a price on big coefficients; for trees, max_depth and min_samples_leaf	`misfit + alpha * size of coefficients`, strength chosen on folds	it helps only if there was variance
L2 and L1	L2 shrinks every coefficient; L1 sets weak ones to zero	squared sizes; absolute sizes	with twin columns L1 keeps one almost by chance
C in LogisticRegression	the brake knob, inverted	`C = 1 / alpha`; small C = strong brake	C = 100 switches the brake off
K-fold cross-validation	k exams; every row sits one; a Pipeline fits each turn	report the mean and the spread	a scaler or selector fitted before the folds

Term	Plain meaning	Rule or formula	Trap
Random forest and boosting	many trees vote; small trees in a chain fix errors	lowers variance; earns the last ranking points	still overfit; cost more to explain
k-means	groups by distance, no answer column	scale first, choose k by use	it never predicts
The two tests	is the gap real; is it worth it	wins fold after fold; `low-end gain x margin > extra cost`	deciding on the mean or the best fold

When to use what. Read training and validation together with the baseline; brake a wide gap and add capacity to two close, high curves; cross-validate whenever a decision rests on a gap; cut folds by time when the model will score the future; ship the simpler model unless the complex one wins fold after fold and its low-end gain clears its cost.

Everything so far, one line each

When	What it gave you
Week 1	Python through data: the revenue tree, descriptive statistics, profile and reconcile before trusting, real or noise, the one-page note
Week 2	SQL and pandas: queries, joins that keep booked and collected apart, window functions, the customer table, Excel for the leadership deck
Build 1	Kalpa Health: profile, clean, reconcile, and the headline claim
Week 4	analyst craft: the metric and its gaming, one slide for three people, lift beyond chance, cohorts and funnels, a forecast against a hard baseline
W5 Mon	frame the prediction, split it, beat 88.7% for calling nobody
W5 Tue	the metric trap: four boxes, the threshold from Rs 40 against Rs 1,200, the budget
W5 Wed	only what is known before the call: `duration` out, AUC 0.93 back to about 0.80
W5 Thu	generalise or memorise: the gap, the brake, five exams, two tests

Three that stick

- Airbnb: a learned vector per listing "mostly led to overfitting", since a listing is booked at most 365 times a year (arXiv 1810.09591, verified 28 October 2026).
- Google Flu Trends chose from 50 million search terms to fit 1,152 points and kept high school basketball terms (Science 2014, verified 28 October 2026).
- A pneumonia model learned that asthma lowers risk; the readable model let doctors see and fix it (Caruana and others, KDD 2015, verified 28 October 2026).